Wangyang Han

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EDUCATION

Shanghai Jiao Tong University *Master of Integrated Circuit Science and Engineering* Sep. 2026 – Present

- **Relevant Coursework:** Advanced Computer System Architecture, SoC Design Methodology, Advanced Digital Integrated Circuit Design, SystemVerilog Design and Verification Methodology

Shanghai Jiao Tong University *Bachelor of Microelectronics Science and Engineering* Sep. 2022 – Jun. 2026

- **GPA:** 91.46/100, **Class Rank:** 2/71
- **Relevant Coursework:** Digital Integrated Circuit Design, Advanced SoC Design, Embedded Systems, Computer Architecture, Artificial Intelligence Hardware Practice

INTERNSHIP

AI Chip Architecture Simulator and Firmware, ByteDance Jan. 2026 – Jun. 2026

Intern, AI Chip Architecture Simulator and Firmware Department

- Adapted GPGPU-Sim's warp scheduler, ALU, LSU, and VRF to the team's proposed architecture; evaluated performance with Softmax, MatMul, and SwiGLU workloads.
- Contributed to C++ algorithm and data-structure optimizations for an AI chip functional model, reducing end-to-end runtime by approximately **8%** on selected critical paths.
- Produced **4 technical reports** on FlashAttention, RDMA, GPU architecture, and NPU architecture.
- Maintained GitLab CI scripts in shell, Python, and C++; synchronized operator-level functional tests for the simulator.

RESEARCH

High-Performance RISC-V SoC Design and Verification Nov. 2025 – Jun. 2026

Undergraduate Thesis Project

- Integrated **UCIe chiplet interface IP**, AXI/AHB/APB buses, memory, and peripherals into a C908-based RISC-V SoC.
- Implemented a **512-bit** AXI-to-UCIe adapter for address mapping and data path conversion.
- Achieved **500 MHz** timing requirements based on **TSMC 28 nm** technology; configured **64 UCIe lanes** at **4 GT/s** each, providing a theoretical peak bandwidth of **256 Gbps** per direction.
- Verified cross-chiplet communication using RTL simulation and C tests; measured RDI-to-RDI latency of about **22 ns**.

CNN Hardware Accelerator Design Nov. 2024 – Nov. 2025

Undergraduate Research Project

- Implemented an INT8 MAC-array CNN accelerator from RTL design and verification to physical implementation.
- Performed synthesis and timing analysis using Synopsys tools and **SMIC 0.18 um** technology.
- Achieved **200,000 samples/s**, equivalent to approximately **49 GMAC/s** (**98 INT8 GOPS**), at 96.7 MHz and 1.1 W.

PROJECT

AI Chip-Based Edge Application Development: Smart Teaching Assistant Jan. 2025 – Aug. 2025

National Undergraduate Integrated Circuit Innovation and Entrepreneurship Competition

- Trained and quantized YOLOv5s and ResNet34 for EIC7700X deployment; added Qwen2-based voice and text interaction.
- Achieved **73.3%** test accuracy with ResNet34 (FER2013) and **73.31%** mAP@0.5 with YOLOv5s (WiderFace).
- Measured inference latencies of approximately **1 ms** for ResNet34 and **3 ms** for YOLOv5s using ES_Run_Model.
- Received First Prize in National Finals.

HONORS

- First Prize (National Finals), National Undergraduate IC Innovation and Entrepreneurship Competition, 2025.
- B-level and C-level Undergraduate Scholarships, Shanghai Jiao Tong University, 2022–2024.

SKILLS

- **Programming:** Familiar with C/C++, Python, and shell scripting; basic knowledge of CUDA kernel programming.
- **Architecture & Systems:** Familiar with GPU architecture, GPGPU-Sim, and performance evaluation; basic knowledge of operating systems, NPU architecture, FlashAttention, RDMA, and model quantization.
- **Hardware:** Proficient in Verilog/SystemVerilog for RTL design and verification; familiar with FPGA programming.
- **Tools & Environments:** Familiar with Windows, Linux, and GitLab CI; basic knowledge of PyTorch.